



Gas Pressure Regulator RB 3200

3300992801
AC

Instruction Manual

Keep this manual easily accessible for all users.

Important safety instructions:

- Only use for dry and clean gases. For use with aggressive gases please contact ITRON for special versions.

Never use with oxygen: Danger of explosion!

- Please respect all national standards and codes of practices for installation, operation, testing and service of gas regulators and of gas pressure regulating stations.
- Before mounting please check the regulator for possible damage due to transport. The sealing surfaces must be clean.
- Arrange enough room for the serving of the regulator.
- To protect the regulator from dirt a sieve or a filter is recommended.
- The regulator must be installed tension free in the piping.
- After installation please check the tightness of the thread connection.
- The technical data given on the name plate must be respected. If necessary safety devices must be installed.
- Repairs and maintenance must be done by trained or qualified personal. Afterwards a tightness test with 1.1 x PS must be performed. When changing pressure containing parts their compliance with the PED must be assured.
- If used with wet gas, internal and external effect of corrosion has to be checked regularly and in case of severe corrosion, the regulator has to be replaced.
- Please don't use solvent or alcohol containing products to clean the regulator.
- Before removing the regulator please depressurise completely. Some gas may still be inside the regulator and the pipe, therefore sufficient ventilation is required.

Remarks for use in potentially explosive atmospheres (ATEX):

- If film of rust is possible (flying rust in the immediate regulator surroundings), all outer aluminium parts have to be protected accordingly (e.g. by varnish).
- The product must be electrically connected to adequately earthed installation.
- For installation, removing or repair of the regulator on site, only tools allowed for the corresponding explosion area can be used.
- The regulator shall not be exposed to: flames, ionised radiation and ultrasound.

Technical Features

Max inlet pressure	10 bar
Outlet pressure	7 mbar – 550 mbar
Operating temperature	-20°C to +60°C
Ambient temperature	-30°C to +60°C (body material)
Acceptable gases	natural gas, town gas, propane, butane, dry air, nitrogen or other non-corrosive gases
Installation	horizontal or vertical
Safety devices	standard relief valve
	Optional built in safety shut-off valve: • Over-pressure shut-off (OPSO) • Under-pressure shut-off (UPS0)
Sensing line	external
Size	1" x 1 1/2"
Connectors	parallel internal thread according to ISO 228/1 or ISO 7/1

Regulator Operation

Regulator Operating Principle

The fluid coming from the upstream pipe enters the inlet chamber (4), passes through the opening of the valve (3) and then expands in the outlet chamber (18) flowing into the downstream pipe.

The downstream pressure, which from the chamber (18) through the pipe (17) enters the regulation diaphragm housing (14), exerts a force on the surface of the diaphragm (9) balancing the load of the calibration spring (12), enabling the valve plug (5) to find an optimum position, ensuring good regulation and the required flow rate.

The series 3200 pressure regulator is fitted with a balanced compensation system, which allows constant downstream pressure regardless of variations of upstream pressure. If during the operation the flow rate increases, due to a demand, or the inlet pressure decreases, the pressure values in the chambers 18 and 14 will tend to decrease. Consequently the force of the spring (12) moves the diaphragm assembly (15) downwards and by means of the lever system (16) adjusts the position of the valve plug (5), thus resetting the pressure and the flow rate to the required values. The reverse occurs when the flow rate demand decreases.

At zero flow, rate the regulator guarantees full tightness.

The regulator is also fitted with a relief valve built into the diaphragm bolder assembly (15) which,

calibrated by means of the spring (13), serves to discharge externally, through the vent (10), possible over-pressure created in the chambers 14 and 18.

The capacity of this valve is limited.

Regulator Calibration

The regulator is supplied with calibration values specified in the order. If a different outlet pressure is required, select the correct spring to obtain this set pressure.

Once the suitable spring (12) is installed, proceed as follows:

- **to increase the outlet pressure value:**
rotate in a clockwise direction the screwing ring (11) until the required value is read on the downstream manometer;
- **to reduce the outlet pressure value:**
proceed likewise but rotate the screwing ring in an anticlockwise direction.

It is recommended to proceed slowly waiting for the pressure to stabilize after each movement of the screwing ring.

These operations can be done either with the regulator in operation or with the downstream stop valve closed (ensuring a small vent is open downstream to give a small flow).

Shut-off Valve Operation

Operating Principle of the Shut-off Valve

The pressure regulator can be fitted with an independent shut-off valve consisting of a shut-off valve casing (21) and a diaphragm assembly (29), which controls a mechanical ball system enabling the release of the valve.

The shut-off valve is triggered when the pressure value in the chamber (22) increases or decreases beyond the calibration values established and thus moves the entire assembly (29) from a balanced position. The stem (27), which is connected to the valve plug (19) is triggered under the force of the spring and comes immediately into contact with the orifice (2), cutting automatically the flow of the fluid.

For series 3211 the shut-off valve operates only for over-pressure shutoff; for series 3212 it operates for both over and low-pressure shut-off.

Calibration of the Shut-off Valve

If the set values of the shut-off are to be modified, proceed as follows:

(A) over-pressure shut-off operation:

rotate the outside shut-off screwing ring in a clockwise direction to increase the pressure values, or anticlockwise to reduce it

(B) low-pressure shut-off operation:

likewise but on the central shut-off screwing ring.

Resetting of the Shut off Valve

The shut-off valve must be reset only after verifying the reasons for the intervention and after having re-established normal operating conditions, proceeding as follows:

- (A) close the downstream stop valve;
- (B) remove the cover (25), unscrewing it (see fig. B1);
- (C) screw it upside down into the stem (23), until it comes into contact with the cover (see fig. B2);
- (D) continue to screw it slowly; this causes the valve to open slowly and pressure fills the downstream pipe piece;
- (E) pull the cover outwards to permit seating of the balls on the stem: this operation ensures resetting (see fig. B3);

After these operations, screw the cover to its housing (see fig. B4) and open slowly the downstream valve.

Caution: for safety reasons, the cover must absolutely be screwed to its housing, as shown in the schematic section.

Installation Procedure

Before installing the regulator it is important to

- (A) clean the upstream pipe; it is recommended to install a cartridge filter upstream of the apparatus.
- (B) check that the regulator has not been visibly damaged during transport.

When installing the regulator

- (C) check that the flow of the gas corresponds to the arrow moulded on the body of the regulator.
- (D) to obtain proper operation of the regulator, **it is essential** for its impulse intake to be connected on the downstream pipe; the pressure intake, according to the installation scheme, must be connected to a straight piece of the downstream pipe.

For this purpose **it is recommended:** 1) to weld the pressure intake onto the upper part of the downstream pipe to avoid impurities; 2) to avoid any points on the impulse line where impurities could collect

Caution: in addition, the regulator must be installed according to good engineering practice. The installation must also comply with national and/or international regulations and standards

Start-up Instruction

Caution: before start-up, the installation must be checked for leaks.

After installing the regulator, check that:

- (1) the stop valves (1), downstream (11) and the bleed cock (9) are closed;
- (2) the gas pressure is not higher than the allowable limits;
- (3) if the apparatus is fitted with a shut-off valve, the shut-off valve must be closed;

After verifying the above mentioned points, proceed as follows:

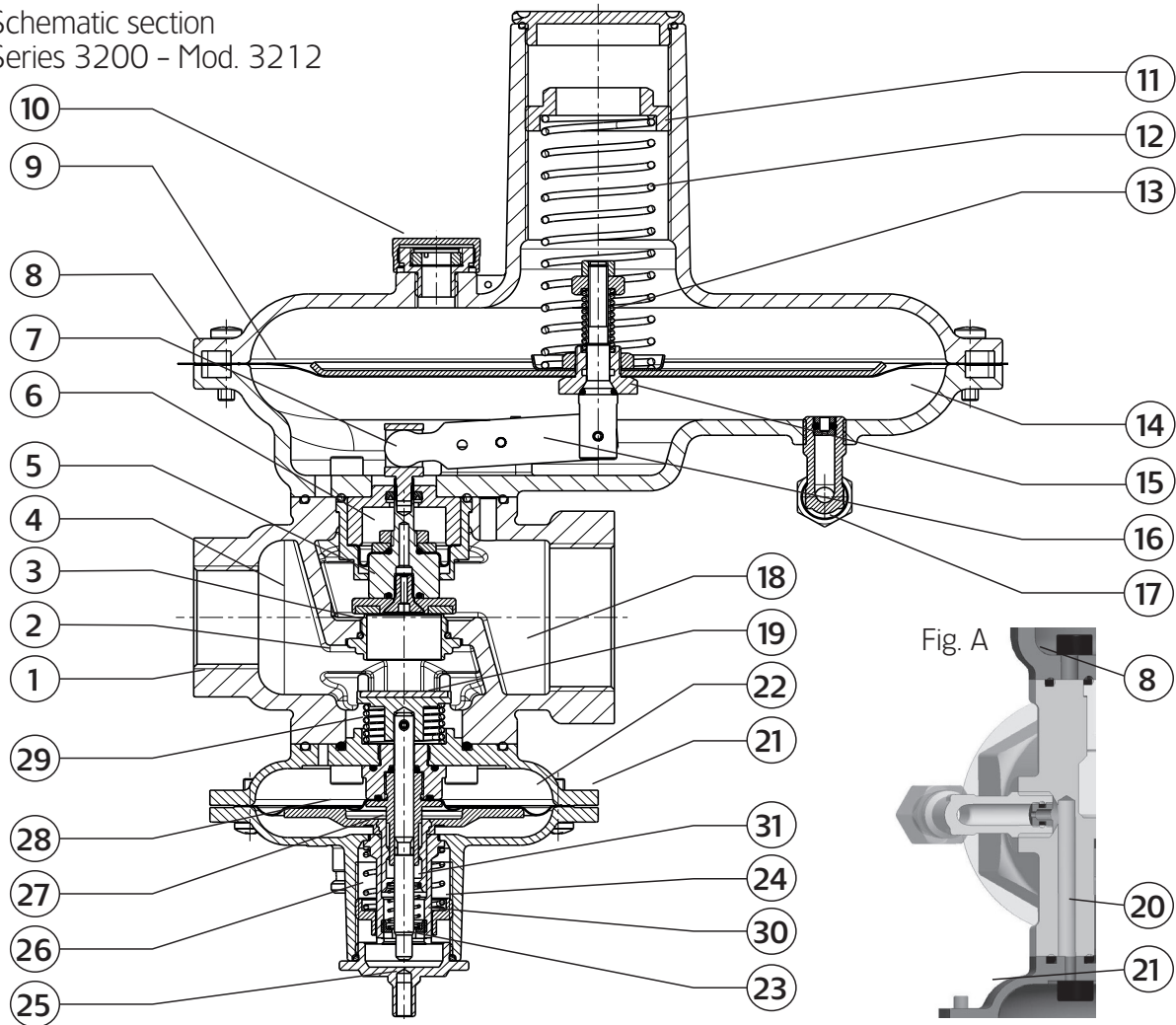
- (A) open the upstream stop valves (1) slowly and just enough to ensure a very small flow of gas; check the upstream pressure gauge (3);
- (B) reset the shut-off valve (see section "Resetting the shut-off valve");
- (C) check that the downstream pressure is rising slowly on the manometer (10): the downstream pressure should stop at the required pressure value or at a value a little higher;
- (D) when the upstream pressure is also stabilised, slowly open the upstream stop valve (1) to a full open position;
- (E) then slowly open the downstream stop valve (11). At this point the regulator is in operation.

Likewise, the process can be followed for the installation of a monitor regulator (4) in line with the working regulator (6) (see installation scheme below), keeping in mind that the manometer (5) installed between the regulators must indicate the same, or little higher, pressure value as the upstream manometer (10).

Annex

RB 3200

Schematic section
Series 3200 - Mod. 3212



Item	Description	Item	Description
1	Body	17	Regulator sensing line port (Fig. A)
2	Filter	18	Outlet Chamber
3	Valve Seat	19	Shut-off Valve Plug
4	Inlet Chamber	20	Shut-off valve sensing line (Fig. A)
5	Valve Plug	21	Shut-off Valve Casing
6	Balancing Chamber	22	Shut-off Control Chamber
7	Drive Shaft	23	Stem
8	Cover	24	Over-pressure Shut-off screwing ring
9	Diaphragm	25	Cover
10	Vent	26	Over-pressure Shut-off Valve Spring
11	Screwing Ring	27	Shut-off Diaphragm Assembly
12	Calibration Spring	28	Safety Shut-off Valve Diaphragm
13	Safety Relief Spring	29	Shut-off Valve Closing Spring
14	Regulation Control Chamber	30	Under-pressure Shut-off Screwing Ring
15	Diaphragm Holder Assembly	31	Under-pressure Shut-off Valve Spring
16	Control Lever		

Resetting the Safety Shut-off Valve

Fig. B1, B2, B3, B4

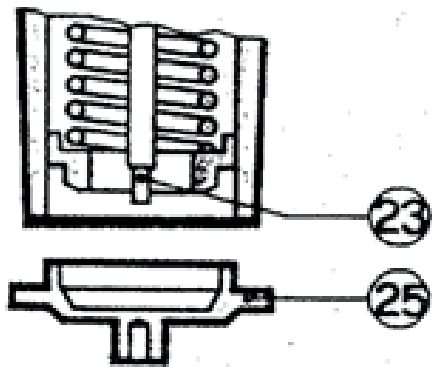


Fig. B1

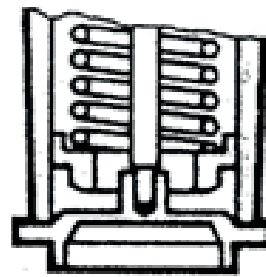


Fig. B2

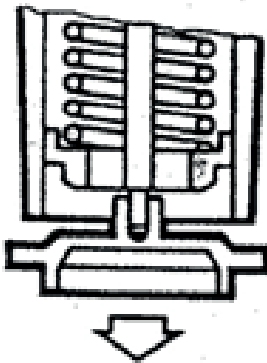


Fig. B3

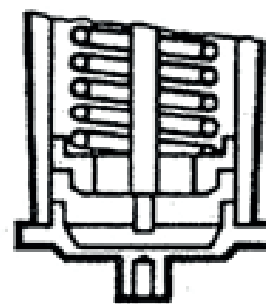


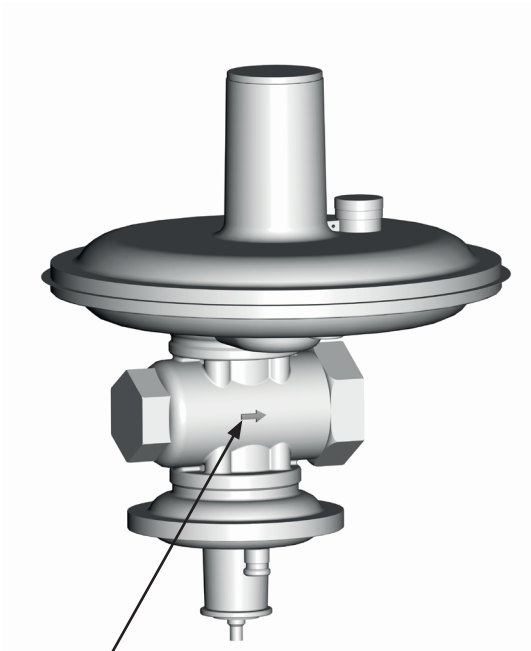
Fig. B4

Table: Safety valve position sensing

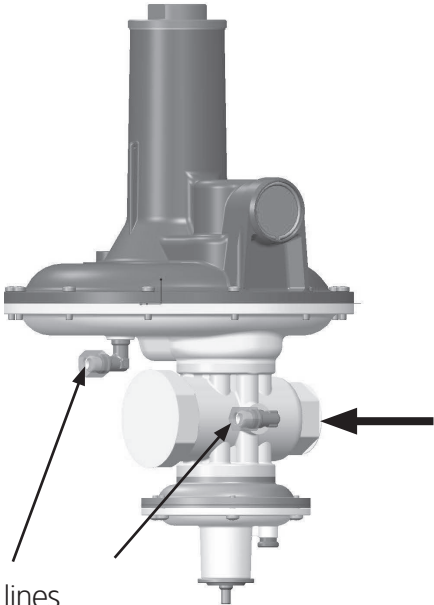
Characteristics	Installation non ATEX	Installation ATEX
Contact power rating	Max. 10 W	Max. 120 mW
Voltage (Break-down)	Max. 180 Vdc	Max. 15 Vdc
Current (Switching)	Max. 50 mA	Max. 50 mA
Capacitance and Inductance	C = 0F, L = 0H	C = 0F, L = 0H
t Operation temperature range	-25°C to +55°C	-25°C to +55°C

The device is a simple apparatus as defined in EN 60079-11 §5.7 and is free to be used in ATEX areas.

Connection



arrow moulded on the body



sensing lines

Outlet Pressure Range

Regulator

	Spring Characteristic				Spring range	
Spring code	D (mm)	De (mm)	Lo (mm)	Lt	LP (mbar)	MP (mbar)
20565166	1,8	35	155	10	7-12	
20565168	2,2	35	155	13	14-25	
20565155	2,7	35	120	11	20-50	
20565156	3	35	120	11,5	28-70	
20565150	3,5	35	100	9,5	20-110	
20565151	4	35	100	10,8	30-160	
955-200-16	4,5	53	207	13		100-140
955-200-17	5,6	55,1	204	13		140-300
955-200-78	6,5	55,6	193	11,5		300-550
955-203-95	5,8	54,6	201	11,4		140-370

Safety Shut-off Valves

Over-Pressure-Shut-off Springs (OPSO)

	Spring Characteristic				Spring range	
Spring code / Code du ressort	D (mm)	De (mm)	Lo (mm)	Lt	std (mbar)	High- pressure (mbar)
20563022	1,5	25	35	5,5	28-60	
20563023	1,7	25	35	5,5	45-90	90-130
20563014	1,9	25	35	5,5	70-130	120-180
20563124	2,2	25	35	5,5	120-220	160-330
20563121	2,5	25	35	5,5	220-400	300-550
20563115	3	25	35	5,5		580-800

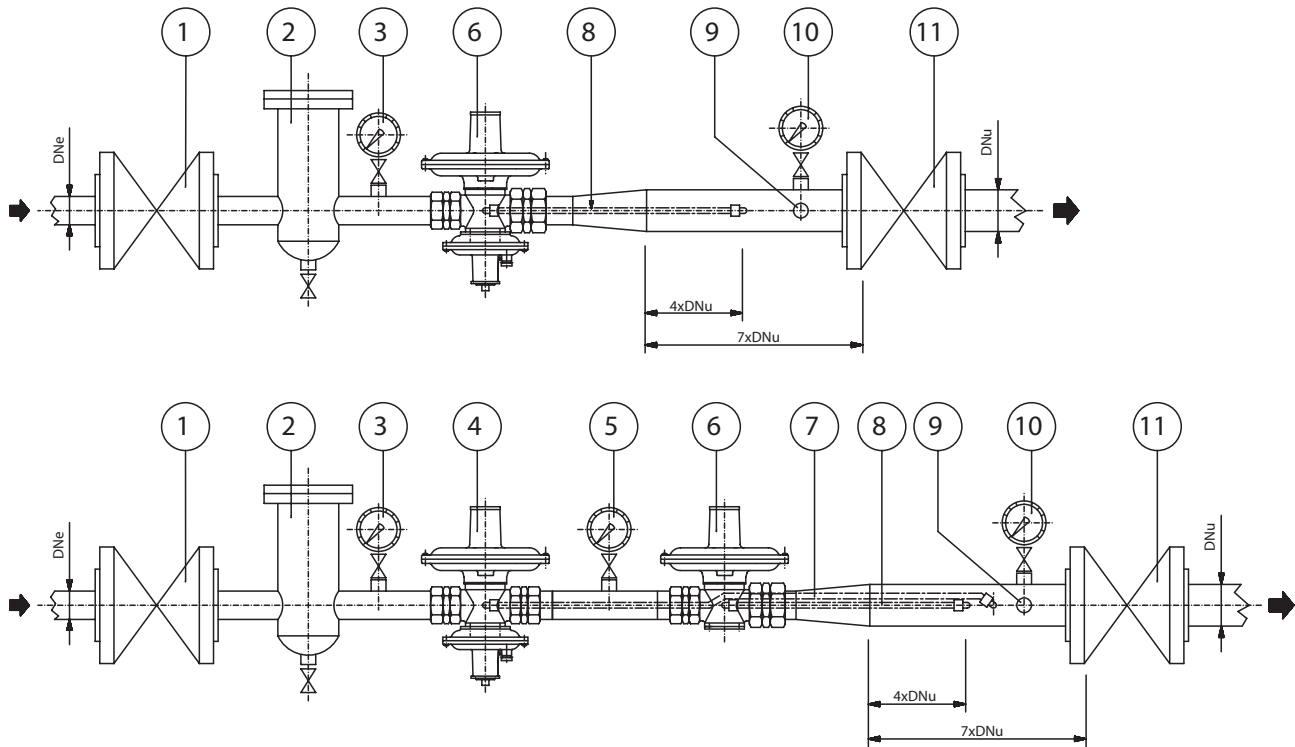
Under-Pressure-Shut-off Springs (UPSO)

	Spring Characteristic / Caractéristiques du ressort				Spring range / Plage du ressort	
Spring code / Code du ressort	D (mm)	De (mm)	Lo (mm)	Lt	std (mbar)	High- pressure (mbar)
20560511	0,8	10	20	7	5-28	
20560518	0,9	10	30	10	27-43	
20560520	1	10,3	25	7,5	30-70	
20565140	1,1	10	24,2	7,8	65-105	
20560517	1,2	10	30	10	100-125	50-165
20560519	1,4	10	30	10		150-300

Relief Valve

The built in relief valve is set 10 mbar (LP) or 45 mbar (MP) above the outlet pressure setting.

Installation Scheme



Item	Description	Item	Description
1	Upstream Stop Valve	7	Monitor Regulator Impulse
2	Cartridge Filter	8	Regulator Impulse Pipe
3	Manometer	9	Bleed Cock
4	Monitor Regulator	10	Manometer
5	Manometer	11	Downstream Stop Valve
6	Regulator		